

Q1. How are NumPy arrays advantageous over Python lists?

NumPy arrays are faster and more efficient than Python lists. They are specially designed for numerical calculations and handling large amounts of data.

Advantages of NumPy Arrays over Python Lists:

1. NumPy arrays perform calculations faster.
2. They use less memory.
3. They support mathematical operations directly.
4. They can handle large datasets efficiently.
5. They are widely used in Data Science, Machine Learning, and Scientific Computing.

Example:

```
 pythonQuestion.py

import numpy as np

arr = np.array([10, 20, 30, 40])
print(arr * 2)

# Output:
# [20 40 60 80]
```

Q2. What are the steps to create 1D, 2D and 3D arrays?

In NumPy, arrays can be created using the `array()` function. Depending on the number of dimensions, arrays are classified as 1D, 2D, and 3D.

1. Creating a 1D Array: A 1D array contains elements in a single row.

Example:

```
 pythonQuestion.py

import numpy as np

arr = np.array([23, 45, 67, 89, 12])
print(arr)

# Output:
# [23 45 67 89 12]
```

2. Creating a 2D Array: A 2D array contains rows and columns.

Example:

```
 pythonQuestion.py

import numpy as np

arr = np.array([[1, 2, 3], [4, 5, 6]])
print(arr)

# Output:
# [[1 2 3]
#  [4 5 6]]
```

3. Creating a 3D Array: A 3D array contains multiple 2D arrays.

Example:

```
 pythonQuestion.py

import numpy as np

arr = np.array([[[1, 2], [3, 4]], [[5, 6], [7, 8]]])
print(arr)

# Output:
# [[[1 2]
#   [3 4]]
#  [[5 6]
#   [7 8]]]
```

Q3. Differentiate between the following:

A. Difference between Array Indexing and Array Slicing

Array Indexing:

1. It is used to access a single element of an array.
2. It returns only one value.
3. It uses a single index.

Example:

```
pythonQuestion.py

import numpy as np

arr = np.array([44, 55, 35, 22, 11])
print(arr[2])

# Output:
# 35
```

Array Slicing:

1. It is used to access multiple elements of an array.
2. It returns a part of the array.
3. It uses start:end syntax.

Example:

```
pythonQuestion.py

import numpy as np

arr = np.array([90, 39, 78, 56, 12])
print(arr[1:3])

# Output:
# [39 78]
```

B. Difference between Copy and View in NumPy

Copy:

1. It creates a new array.
2. Changes in the copied array do not affect the original array.
3. It uses the copy() method.

Example:

```
pythonQuestion.py

import numpy as np

arr = np.array([7, 14, 21, 28, 35])
new_arr = arr.copy()

print("Original array:", arr)      # [ 7 14 21 28 35]
print("Copied array:", new_arr)    # [ 7 14 21 28 35]
```

View:

1. It creates another view of the same array.
2. Changes in the view also affect the original array.
3. It uses the view() method.

Example:

```
pythonQuestion.py

import numpy as np

arr = np.array([9, 4, 6])
new_arr = arr.view()

print("Original array:", arr)      # [9 4 6]
print("View of the array:", new_arr) # [9 4 6]
```

Q4. What is a Normal Distribution? Mention its characteristics.

Normal Distribution is a type of probability distribution in which most of the data is concentrated around the mean (average) value. It is also called the Bell-Shaped Curve because of its shape.

Characteristics of Normal Distribution:

1. It has a bell-shaped and symmetrical curve.

2. The mean, median, and mode are equal.
3. Most of the values are near the mean.
4. The curve is highest at the center and decreases towards both ends.
5. The total area under the curve is equal to 1.

Example:

The heights of students in a class usually follow a Normal Distribution. Most students have an average height, while very few are extremely tall or extremely short.

Q5. What is a Binomial Distribution? Give one real-life example.

Binomial Distribution is a type of probability distribution that shows the probability of getting a fixed number of successes in a fixed number of trials. Each trial has only two possible outcomes: Success or Failure.

Characteristics of Binomial Distribution:

1. It has a fixed number of trials.
2. Each trial has only two possible outcomes (Success or Failure).
3. The probability of success remains the same in every trial.
4. Each trial is independent of the others.

Example:

Tossing a coin 10 times is a binomial distribution. In each toss, there are only two possible outcomes:

- Head (Success)
- Tail (Failure)

Real-Life Example:

A teacher checks whether students pass or fail in an exam. Each student can have only two outcomes: Pass or Fail.

Q6. Explain the following NumPy concepts with suitable examples:

1. Shape

The shape of a NumPy array tells us the number of rows and columns in the array.

Example:



pythonQuestion.py

```
import numpy as np

arr = np.array([[11, 42, 63], [14, 35, 46]])
print(arr.shape)

# Output:
# (2, 3)
```

2. Reshape

The reshape() function is used to change the shape of an array without changing its data.

Example:



pythonQuestion.py

```
import numpy as np

arr = np.array([6, 2, 6, 1, 9, 6])
new_arr = arr.reshape(2, 3)
print(new_arr)

# Output:
# [[6 2 6]
#  [1 9 6]]
```

3. Join

The join() function is used to combine two or more arrays into one array.

Example:



pythonQuestion.py

```
import numpy as np

arr1 = np.array([11, 45, 33])
arr2 = np.array([90, 54, 56])
arr = np.concatenate((arr1, arr2))
print(arr)

# Output:
# [11 45 33 90 54 56]
```

4. Split

The `split()` function is used to divide an array into multiple smaller arrays.

Example:



pythonQuestion.py

```
import numpy as np

arr = np.array([16, 32, 48, 64, 80, 96])
new_arr = np.array_split(arr, 3)
print(new_arr)

# Output:
# [array([16, 32]), array([48, 64]), array([80, 96])]
```

5. Filter

Filtering is used to select specific elements from an array based on a condition.

Example:

```
pythonQuestion.py

import numpy as np

arr = np.array([10, 20, 30, 40, 50])
new_arr = arr[arr > 25]
print(new_arr)

# Output:
# [30 40 50]
```

Q7. What is Seaborn? How is it useful for visualizing data generated using NumPy?

Seaborn is a Python library used for creating attractive and informative graphs. It is built on top of Matplotlib and makes data visualization simple and effective. Seaborn works well with NumPy arrays and helps display data in graphical form.

Uses of Seaborn with NumPy:

1. It creates beautiful and easy-to-understand graphs.
2. It helps visualize data stored in NumPy arrays.
3. It supports different types of charts such as line plots, bar plots, scatter plots, and histogram plots.
4. It is widely used in Data Analysis and Machine Learning.

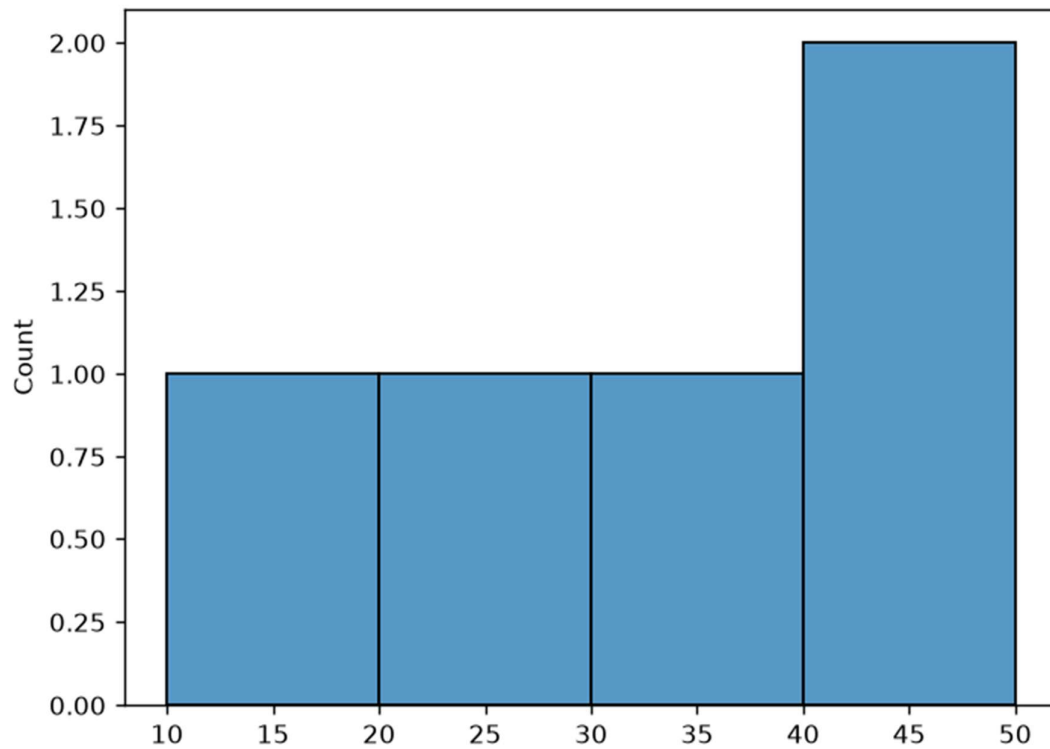
Example:

```
pythonQuestion.py

import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = np.array([10, 20, 30, 40, 50])
sns.histplot(data)
plt.show()
```

Output:



Q8. Explain the difference between Normal Distribution and Binomial Distribution with one real-life example of each.

Normal Distribution:

1. It is a continuous probability distribution.
2. It has a bell-shaped and symmetrical curve.
3. The mean, median, and mode are equal.
4. It is used for continuous data.

Example:

The heights of students in a class usually follow a Normal Distribution because most students have an average height, while only a few are very tall or very short.

Binomial Distribution:

1. It is a discrete probability distribution.
2. It has only two possible outcomes: Success or Failure.
3. It is based on a fixed number of trials.
4. It is used for countable data.

Example:

Tossing a coin 10 times is a Binomial Distribution because each toss has only two possible outcomes: Head or Tail.

Difference:

1. Normal Distribution is used for continuous data, whereas Binomial Distribution is used for discrete data.
2. Normal Distribution has a bell-shaped curve, whereas Binomial Distribution is based on Success or Failure.
3. Normal Distribution is used for measurements like height and weight, whereas Binomial Distribution is used for experiments like coin tosses and pass/fail results.

Q9. Write a Python program to count how many times a given digit appears in a number.



pythonQuestion.py

```
number = input("Enter a number: ")
digit = input("Enter the digit to count: ")

count = 0

for i in number:
    if i == digit:
        count += 1

print("The digit", digit, "appears", count, "times.")

# Output:
# Enter a number: 834797324973294723
# Enter the digit to count: 4
# The digit 4 appears 3 times.
```

Q10. Write a Python program to find the number of days in a given month of a given year.



pythonQuestion.py

```
year = int(input("Enter year: "))
month = int(input("Enter month (1-12): "))

if month == 2:
    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
        print("Number of days = 29")
    else:
        print("Number of days = 28")

elif month in [4, 6, 9, 11]:
    print("Number of days = 30")

else:
    print("Number of days = 31")

# Output:
# Enter year: 2020
# Enter month (1-12): 2
# Number of days = 29
```